The empirical recurrence for OEIS A282590: recovering a conjecture that is not written down

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Abstract

OEIS A282590 records "Empirical recurrence of order 93", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 260$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 93, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 93 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282590 is "Number of n X 5 0..1 arrays with no 1 equal to more than two of its king-move neighbors, with the exception of exactly two elements.". It has offset 1 and begins

$$0, 52, 4083, 73469, 1551972, 30003220, 546521435, 9746960274, \ \ldots$$

The entry states:

> Empirical recurrence of order 93 (see link above).

As of the "Last modified" line on the live entry (Oct 20 2022 23:40:13, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1811$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 260$ of the 1811, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 260$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 520$ exact terms of the model returns order 93 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{93} c_i\, a(n-i),$$

c_1	42	c_{25}	−3946820121128307	c_{49}	4546645275975745248	c_{73}	−88378334409591
c_2	−555	c_{26}	5666938331520483	c_{50}	−264609110572490750109	c_{74}	−37004073221470
c_3	1853	c_{27}	−10157498995242882	c_{51}	−927287931879957749287	c_{75}	27090151270238
c_4	−1101	c_{28}	3653537185224894	c_{52}	−160988939992608811572	c_{76}	−9064459216358
c_5	201930	c_{29}	14714833154334798	c_{53}	−311109765325814407503	c_{77}	57234468560508
c_6	−1532236	c_{30}	−38724243140196890	c_{54}	989489884367363451392	c_{78}	13248113769226
c_7	1629273	c_{31}	185828625098454612	c_{55}	1175811166417756845015	c_{79}	−7190481459179
c_8	−18216450	c_{32}	−253249371799931472	c_{56}	271747858176731416959	c_{80}	12955638885992
c_9	262415525	c_{33}	437356700255084989	c_{57}	315578559962450779753	c_{81}	2105705267257
c_{10}	−920792628	c_{34}	−836818333589643834	c_{58}	−1333548353825179952379	c_{82}	−203170875603
c_{11}	2601075648	c_{35}	−132449853187379745	c_{59}	−600063894448546822197	c_{83}	851226850815
c_{12}	−17659055274	c_{36}	−457676635336369890	c_{60}	−86554614042937914969	c_{84}	−207377914830
c_{13}	82129250070	c_{37}	−1996562459424670887	c_{61}	−58818659141353416423	c_{85}	71262172068
c_{14}	−298663072989	c_{38}	7780465019848494804	c_{62}	838092074524058211138	c_{86}	153442188444
c_{15}	1072147408644	c_{39}	−2511483158898807642	c_{63}	−122355016291758861459	c_{87}	−57983775742
c_{16}	−3065177710578	c_{40}	21867163063536007137	c_{64}	−612542658151290754989	c_{88}	163817054400
c_{17}	7671643617390	c_{41}	−15013267188400425252	c_{65}	314058144219769659831	c_{89}	−3284552908
c_{18}	−17334961280803	c_{42}	−11145912857894738150	c_{66}	190447258616383341802	c_{90}	3279982848
c_{19}	27365988825072	c_{43}	−45635140924875632400	c_{67}	−193997239072554393930	c_{91}	−94009344
c_{20}	−35085626841696	c_{44}	−85929853074906175887	c_{68}	−912733808548800990	c_{92}	1050624
c_{21}	2054832841973	c_{45}	67028306590097261458	c_{69}	56249775681893592413	c_{93}	−4096
c_{22}	214840874101893	c_{46}	11522951537644798998	c_{70}	−18313526339749431447		
c_{23}	−633339380314410	c_{47}	352827672769269839601	c_{71}	−6033387065946450012		
c_{24}	1820334400986075	c_{48}	125256918506947724020	c_{72}	5328156777366513487		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 93, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $93 + 4$ terms removed returns order 93 as well, so $\deg q_{\min} = 93 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $93 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 93 that this sequence satisfies — and that family turns out to have exactly one member.

References

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